Performance Analysis of CNNs for Intelligent Forest Fire Monitoring

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***Abstract*—Forest fires are among the main environmental disasters worldwide, causing air pollution, biodiversity loss, soil degradation, water contamination, and severe ecological damage. Recent advances in artificial intelligence have enabled the development of automated wildfire detection systems based on image analysis. However, deploying these models on embedded devices remains challenging because of their limited processing power, memory capacity, and storage resources, which may compromise detection performance. The study evaluates a compact convolutional neural network for wildfire image classification. The developed approach is evaluated on the Kaggle wildfire image dataset. Published Picture NN metrics are reported solely as contextual reference because the two systems were not evaluated under an identical experimental protocol. Experimental results demonstrate that the proposed approach achieved an accuracy of 98.14 percent, a recall of 98.52 percent, a precision of 98.04 percent, and an F1-score of 98.28 percent. These results indicate promising classification performance on the evaluated dataset. Inference latency, runtime memory usage, and energy consumption have not been measured on the target embedded hardware; therefore, embedded efficiency and deployment feasibility remain unverified.**

***Index Terms*—Convolutional Neural Network, AI Application, Embedded Machine Learning, Fire Detection**

I. Introduction

Forest and woodland fires are a significant environmental problem, annually degrading large areas of forests and fields worldwide [1]. They can be caused by both human factors, such as land clearing for agriculture and livestock or arson, and natural ones, such as electrical discharges [2]. In Brazil, fires have increased due to deforestation and the expansion of pasture and agricultural areas [3].

According to [4], more than 1.6 million km² were burned in Brazil, with the deforested Amazon area reaching 11,079 km², a 10% increase over 2019. Notably, 65% of this burned area occurred in primary and secondary forests rather than already deforested land, mainly in Mato Grosso, Pará, and Tocantins [5]. Beyond biodiversity loss, these fires contribute to greenhouse gas emissions and degraded air quality and soil fertility [6].

Forest destruction also alters rain cycles and increases temperatures [7], underscoring the need for effective strategies to prevent and control forest fires [8].

Early fire detection allows for a faster, more effective response, minimizing damage to flora and fauna [9], reducing the burned area and associated greenhouse gas emissions [10], and protecting properties and community safety [11].

With the aim of prior identification of fire outbreaks and their subsequent treatment, fire detection systems are developed. Such systems use advanced technologies to identify and monitor the occurrence of these events in forest areas. These technologies include remote sensing systems, such as satellites and drones, as well as high-resolution sensors and cameras [12].

Artificial intelligence (AI) has been increasingly applied to forest fire detection, from artificial neural networks predicting fire occurrence to machine learning algorithms analyzing satellite imagery for early signs of fire [13], as well as embedded systems integrating high-definition cameras and lightweight neural networks for forest fire smoke monitoring [14].

Among these techniques, Convolutional Neural Networks (CNNs) have been extensively used for forest fire detection given their intrinsic ability to automatically learn discriminative features from audio or image data [15].

Wildfire image classification for resource-constrained applications requires balancing predictive performance and model complexity. The specific question investigated here is whether a compact CNN with reduced-resolution inputs can achieve promising Fire/No fire classification performance on the evaluated dataset. Fire CNN addresses this question using $90\times90\times3$ inputs and 49,824 parameters. Published results from an embedded system provide contextual reference, while hardware feasibility and cross-dataset generalization remain outside the demonstrated scope. This gap motivates the present study, whose main contributions are as follows:

- A compact CNN, referred to as Fire CNN, is presented for wildfire image classification using $90 \times 90 \times 3$ inputs and 49,824 parameters.
- Fire CNN results are contextualized using published Picture NN metrics from [16], with differences in evaluation protocols explicitly acknowledged.
- The architecture, hyperparameters, and training protocol of Fire CNN are detailed to support reproducibility.

II. Related Works

Contemporary fire detection and alert systems predominantly rely on satellite surveillance of fire-prone areas [17] or

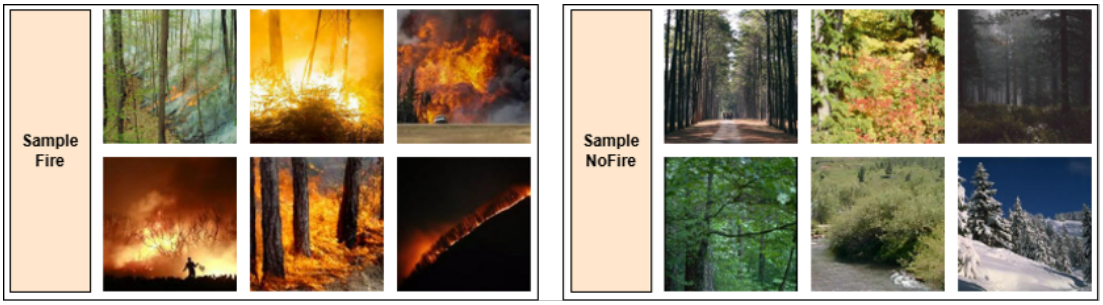


Fig. 1. Representative samples from the wildfire image dataset showing the Fire and No Fire classes.

on the analysis of data collected from strategically positioned local sensors [18]. These data are transmitted to central servers for evaluation or, more recently, processed directly on edge devices [19], enabling local data processing to eventually trigger alarm transmissions.

Several studies have explored the use of deep learning techniques for wildfire detection in images captured from various sources, such as satellites, drones, or fixed cameras. Some of these studies and their main contributions are presented below.

In a comparative study conducted by [20], traditional classifiers and deep neural networks were investigated for wildfire detection. The proposed model, which integrates deep features extracted using the ResNet50 model with the DNN-3 classifier, achieved a classification accuracy of 97.11% in wildfire detection.

The study by [21] presents an automatic alert system for wildfire detection, integrating sensors, drones, and a state-of-the-art deep learning algorithm to monitor and report fires to the relevant authorities in real time.

An innovative wildfire prediction system, named Wildfire Prediction System (WiPreSy), is proposed by [22]. Leveraging Internet of Things (IoT) sensors and deep neural networks, WiPreSy monitors real-time changes in climatic parameters to predict wildfire intensity.

The study in [23] proposes a framework for wildfire inspection using deep convolutional neural networks (CNNs), demonstrating effectiveness in the early identification of such events.

The study by [24] describes an advanced method for rapid localization and segmentation of wildfire core areas in aerial images, employing a saliency detection algorithm and a deep learning architecture called “Fire_Net,” achieving an accuracy of 98%.

In an emerging scenario, [25] discusses solutions for early wildfire detection, integrating Unmanned Aerial Vehicles (UAVs) equipped with specialized cameras and sensor-based systems using LoRaWAN (Long Range Wide Area Network), a low-power, long-range communication protocol designed to connect IoT devices over wide areas.

Additionally, [26] proposes a UAV-based wildfire detection method utilizing a convolutional neural network (CNN) for early fire identification. The study highlights the advantages of UAVs equipped with color and infrared cameras, emphasizing their low cost and high maneuverability. Similarly, [27] presents a comparable approach, reinforcing the practical benefits of UAVs in wildfire detection and suppression.

More recently, [28] proposed HybriDet, a hybrid CNN-Transformer architecture for wildfire detection in remote sensing imagery, and [29] introduced WISE, a lightweight, weakly supervised model for onboard fire and smoke detection and localization. Although these works target object detection and segmentation tasks rather than the binary image classification problem addressed in this study, they illustrate the continuing trend toward combining higher model capacity with resource-aware design for edge deployment.

III. WORK USED AS REFERENCE

The reference study used for contextual comparison is the system proposed in [16], which introduces a wildfire detection framework based on machine learning embedded in a low-power system, using a publicly available dataset suitable for training and inference in other networks.

The reference system is a Video Surveillance Unit (VSU) prototype that monitors, detects, and signals wildfires using two machine learning algorithms deployed on dedicated hardware—one for audio, one for images—and transmits alerts via the LoRaWAN protocol.

The reference system first uses an audio-based model to screen onboard microphone recordings for a potential fire condition; if triggered, an image-based model analyzes photographs from the onboard camera to confirm the event. The fire probabilities inferred from both models are combined into a LoRaWAN packet and transmitted to notify responsible personnel, after which the cycle restarts.

Although both models perform comparably in isolation, their combined use enhances system reliability and overall accuracy, reported at 96.15%, 100.00% precision, 92.30% recall, and 96.00% F1-score.

Fire CNN is evaluated for wildfire image classification. Only the image-based results from [16] are used as contextual reference.

The dataset utilized for training the classification model consists of forest images captured under various climatic conditions (e.g., clear skies, rain, fog), across different seasonal cycles and times of the day, considering diverse lighting conditions. These images are categorized into two classes:

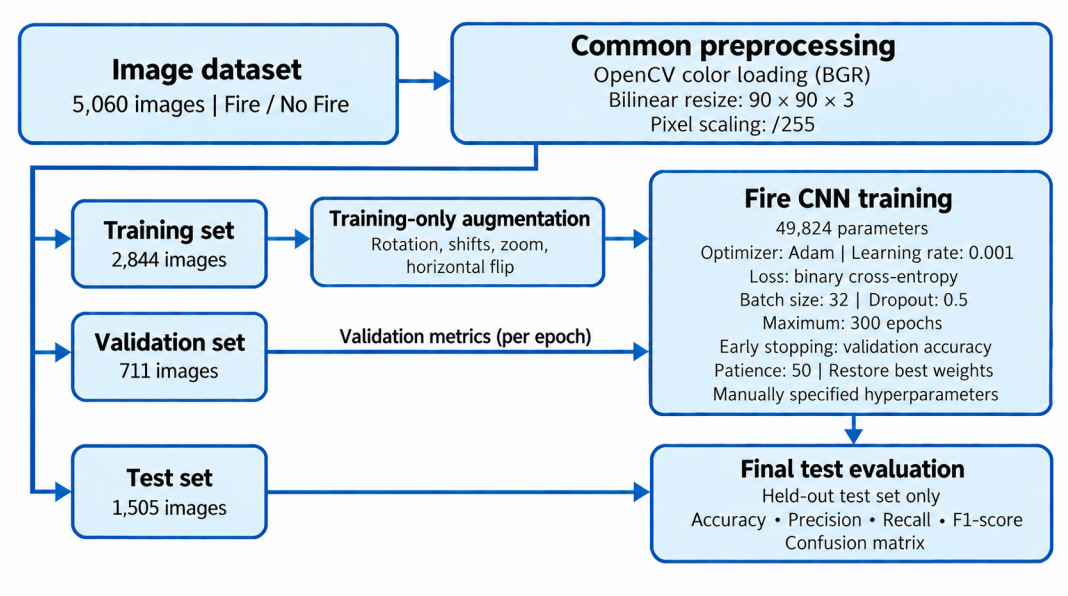


Fig. 2. Overview of the Fire CNN experimental workflow.

Fire, containing forest images with active fires under the various climatic conditions mentioned, and No Fire, containing forest images without any fire occurrence.

The dataset used in the reference study comprises publicly available data from [30]. Fig. 1 presents samples of this dataset with images classified as Fire and No Fire.

In addition to these images contained in the dataset, there are also images captured by the camera built into the prototype. In total, the dataset includes 5,060 images, whose contents (image type) and quantity are presented in Table I. The images captured by the camera on board the prototype had a resolution of 320×240 pixels in grayscale. Additionally, Noise images—inconsistent samples representing irrelevant or undesired information unrelated to the general data pattern—were introduced to ensure the model is trained with relevant, representative data and to minimize interference from unwanted information.

TABLE I
DISTRIBUTION OF THE IMAGE DATASET

Class	Image Type	Number of Samples
Fire	Clear day	1,108
	Night	1,212
No Fire	Snow	389
	Clear day	1,297
	Rain	658
	Noise	396
Total		**5,060**

In the reference study, due to memory constraints imposed by the hardware on which the model was implemented and considering that the prototype's microprocessor is a Cortex M7, which also manages LoRaWAN connectivity when a wildfire is detected, the input image used in the neural network was resized during training. Initially captured at 320 × 240 pixels (native resolution of the integrated camera), the images were downscaled to 96 × 96 pixels.

It is important to highlight that, due to potential distortions in aspect ratio, the images underwent a normalization process. Additionally, given the specific characteristics of the integrated camera, samples were captured in grayscale. This choice facilitated feature extraction, as the distinctive properties could be effectively represented by the grayscale intensity levels of each pixel in the sample.

This grayscale preprocessing was adopted in the reference study to accommodate the characteristics of the prototype's camera. The preprocessing used for the proposed Fire CNN is described separately in Section IV.

For image classification, the study selected MobileNetV2, an open-source CNN developed by Google that significantly reduces the number of parameters compared to conventional CNNs while remaining efficient on embedded devices; in [16], this network is referred to as "Picture NN."

As a pretrained model, MobileNetV2 leverages transfer learning from large-scale image datasets, which proved sufficient for the VSU prototype and significantly reduced training time compared to full-network training, while retaining strong generalization capability.

For Picture NN, the reference study [16] reports 84.60% accuracy, 100.00% recall, 76.50% precision, and an F1-score of 86.70%. These published values are included in Table III for contextual comparison.

IV. MATERIALS AND METHODS

The CNN training and evaluation experiments were conducted using Google Colaboratory (Google Colab), in a cloud-hosted virtual machine equipped with an Intel Xeon CPU operating at 2.20 GHz and 12.67 GiB of system RAM. The

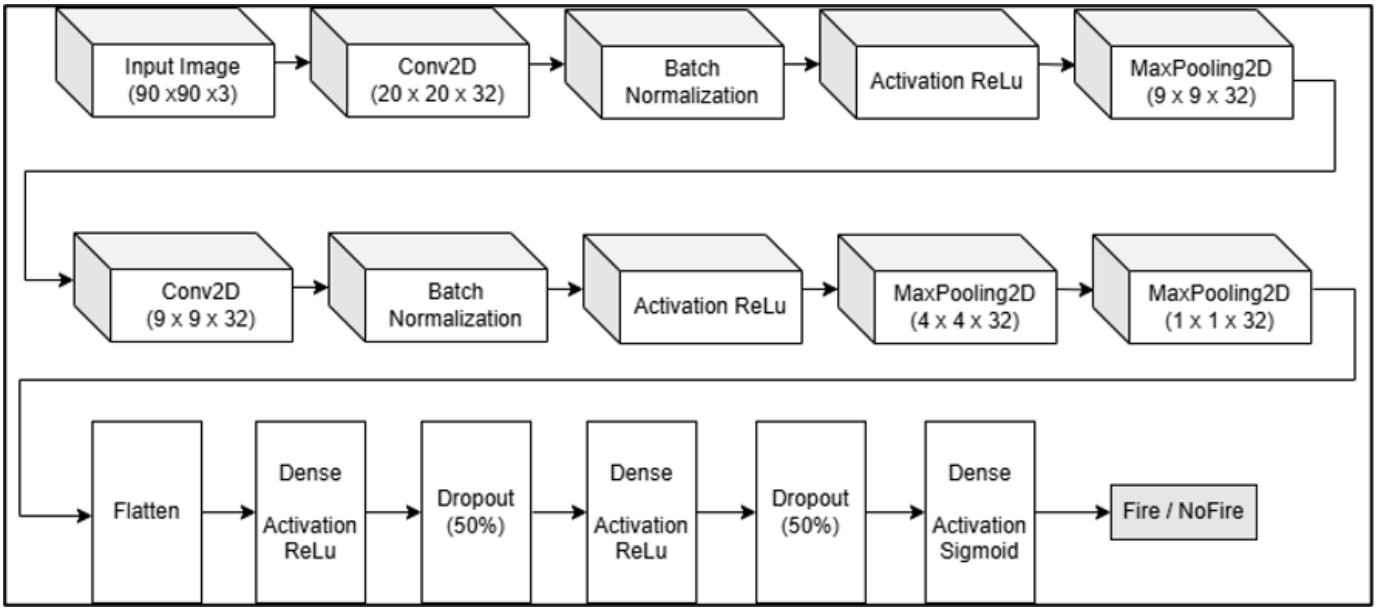


Fig. 3. Architecture of the proposed convolutional neural network for wildfire image classification

environment ran Linux (kernel version `6.6.122+`, `x86_64` architecture). Computations were performed on the CPU, as no GPU was available to TensorFlow in the experimental session. The model was implemented in Python using TensorFlow and Keras, with NumPy, Scikit-learn, and OpenCV supporting data preprocessing and performance evaluation.

Fire CNN uses 90×90 input images to limit spatial computational demands while maintaining a compact architecture. With the specified convolution and pooling operations, this input produces a final feature map of $1 \times 1 \times 32$, yielding only 32 inputs to the dense layers. Images were resized directly using bilinear interpolation, which may alter their original aspect ratio and suppress fine details, including small or distant flames. This resolution represents a design choice; its optimality was not established through comparisons with alternative resolutions.

Images were loaded using OpenCV's `cv2.imread` with its default color-loading mode, producing three-channel arrays in BGR order. No explicit grayscale conversion was applied. The resizing described above used bilinear interpolation, resulting in input tensors of size $90 \times 90 \times 3$. Pixel values were divided by 255 to scale them to $[0, 1]$. When a source image is grayscale, the default loading mode represents its intensity values across three channels, without introducing additional color information.

Fig. 2 summarizes data preparation, CNN training with validation-accuracy monitoring, and final evaluation on the held-out test set.

Fig. 3 presents a more detailed view of the convolutional neural network architecture proposed in this work, which will be further explained in the following sections.

The proposed CNN model follows a sequential architecture composed of multiple distinct layers. The first convolutional layer uses 32 filters of size 11×11, a stride of 4, valid padding, and no bias. Batch normalization is applied next, followed by the ReLU (Rectified Linear Unit) activation function. Next, a max pooling layer (MaxPooling2D) is employed to reduce spatial dimensionality, followed by a second convolutional layer, batch normalization, and activation. Two additional max pooling layers are incorporated, culminating in a flatten layer, which transforms the data into a one-dimensional format.

The dense part of the architecture begins with a fully connected layer containing 128 units (neurons), followed by a dropout layer to reduce the risk of overfitting, which occurs when the model becomes excessively tailored to the training data. Another dense layer with 64 units is incorporated, followed by a second dropout layer. The final layer is also a dense layer with a single output unit, indicating the binary nature of the classification problem addressed in this study. The total number of parameters in the network is 49,824, of which 49,696 are trainable.

Fire CNN combines convolution, normalization, and pooling to extract discriminative features, with the final dense layers performing the Fire/No Fire classification decision.

To train the CNN, careful selection of various parameters was performed. Initially, the optimizer, loss function, and evaluation metrics were specified. The Adam optimizer was chosen to refine the network weights, while binary cross-entropy was employed as the loss function, given its suitability for binary classification tasks. The primary metric monitored during training is accuracy.

To control the training process and prevent overfitting, an early stopping mechanism was implemented. This mechanism monitors the accuracy metric, waiting for 50 consecutive epochs without improvement before terminating training. Additionally, the restore_best_weights option is enabled to revert

TABLE II
LAYER CONFIGURATION OF THE PROPOSED MODEL

Name	Layer (type)	Output shape	Parameters
Input	Input layer	$90 \times 90 \times 3$	0
Conv2D	Convolution	$20 \times 20 \times 32$	11616
BN	Batch Normalization	$20 \times 20 \times 32$	128
AC	Activation	$20 \times 20 \times 32$	0
Max pooling	Pooling, size $= 3 \times 3$	$9 \times 9 \times 32$	0
Conv2D_1	Convolution	$9 \times 9 \times 32$	25600
BN	Batch Normalization	$9 \times 9 \times 32$	128
AC	Activation	$9 \times 9 \times 32$	0
Max pooling_1	Pooling, size $= 3 \times 3$	$4 \times 4 \times 32$	0
Max pooling_2	Pooling, size $= 3 \times 3$	$1 \times 1 \times 32$	0
Flatten	Flatten	32	0
Fully connected	Dense (128 neurons)	128	4096
Dropout	Dropout (0.5)	128	0
Fully connected_1	Dense (64 neurons)	64	8192
Dropout_1	Dropout (0.5)	64	0
Fully connected_2	Dense (1 neuron)	1	64

the model's weights to those that achieved the highest accuracy during training.

Training batches were generated using `ImageDataGenerator.flow` with a batch size of 32. This value influences both training efficiency and computational resource consumption. Training was performed for a maximum of 300 epochs, with early stopping.

The dataset contained 5,060 images, partitioned into 2,844 training, 711 validation, and 1,505 test samples. The dataset includes images with varying lighting conditions and smoke densities, as well as scenes containing potential false-positive triggers, such as sunsets and fog. The training–validation pool of 3,555 images was randomly split in an 80:20 ratio, while the test set was loaded separately. A duplicate-image audit confirmed that no identical images were shared across the training, validation, and test sets. Data augmentation was applied exclusively to training images using `datagen.flow`. Fig. 4 reports predictions exclusively on the 1,505 test images.

Table II summarizes the Fire CNN architecture, comprising 49,824 parameters. This parameter count characterizes model compactness but does not establish embedded efficiency. Target-device measurements of inference latency, peak RAM usage, and flash occupancy remain necessary. Hyperparameters were manually specified, without a systematic search. The evaluated configuration used Adam, a learning rate of 0.001, binary cross-entropy, and a dropout rate of 0.5. Training was limited to 300 epochs, with early stopping monitoring validation accuracy and a patience of 50 epochs. This configuration is reported as the evaluated setting, rather than an experimentally established optimum.

Data shuffling was applied before each epoch, a learning rate (LR) of 0.001 was used as one of the key hyperparameters, and a 50% dropout rate was applied to prevent overfitting, consistent with Table II and the early stopping procedure described above.

V. RESULTS

After training the previously described convolutional neural network using the specified dataset, the proposed model achieved the performance metrics presented in Table III, shown below.

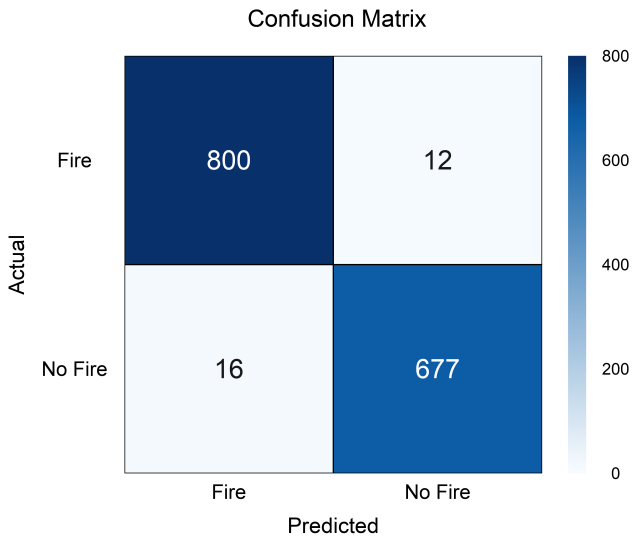


Fig. 4. Confusion matrix of the proposed convolutional neural network

TABLE III
FIRE CNN RESULTS AND PUBLISHED PICTURE NN METRICS

Model	Accuracy	Recall	Precision	F1-score
Picture NN	84.60%	100.00%	76.50%	86.70%
Fire CNN	98.14%	98.52%	98.04%	98.28%

Fire CNN achieved an accuracy of 98.14%, a recall of 98.52%, a precision of 98.04%, and an F1-score of 98.28% on the evaluated test set. Table III also presents values reported for Picture NN as contextual information. Because the evaluation protocols differ, these values do not establish the superiority of either model.

Fig. 4 presents the confusion matrix of the proposed CNN. The following details are noted: 800 images were correctly classified as Fire (true positives, TP); 677 images were correctly classified as No Fire (true negatives, TN); 12 images with an actual wildfire were missed by the model (false negatives, FN); and 16 images without a wildfire were incorrectly flagged as Fire (false positives, FP).

VI. DISCUSSION

The results indicate promising classification performance on the evaluated dataset. However, differences in test data and decision protocols prevent attributing the numerical differences from Picture NN solely to the architecture. Recent approaches discussed in Section II also address different tasks and deployment settings. A controlled benchmark would therefore be required to establish comparative advantages over these methods.

The confusion matrix (Fig. 4) confirms this performance, with few false positives and false negatives, though further improvements in wildfire image classification remain possible.

Architectural compactness does not establish embedded efficiency. Inference latency, peak runtime memory usage, and energy consumption were not measured on the target device and remain subjects for future validation.

VII. CONCLUSIONS

Fire CNN was evaluated for wildfire image classification, with published results from [16] providing contextual reference.

The analysis focused on the algorithm responsible for processing the image dataset, which contains photographs of forests under various climatic conditions and seasons. The reference study used a MobileNetV2-based image classifier [16], whose published classification metrics provide contextual information for the present analysis.

Fire CNN combines a compact architecture with promising performance on the evaluated wildfire image dataset. The reported results support its further investigation, but do not establish superiority over the reference system or recent state-of-the-art methods. Generalization and embedded deployment require additional validation.

The proposed architecture showed promising results for wildfire detection from forest images on the evaluated dataset. Broader claims of superiority over the state of the art would require comparison against additional baselines and datasets beyond the single reference system and Kaggle dataset used here, including recent lightweight and hybrid architectures such as those in [28] and [29]. Furthermore, it is essential to consider the implementation context, available resources, and specific application priorities when selecting a CNN model.

Future work will evaluate Fire CNN on additional datasets and compare it with lightweight CNN baselines under a common evaluation protocol. Robustness tests will assess the effects of lighting, smoke density, and potential false-positive triggers, such as sunsets and fog. Deployment on embedded hardware will enable measurements of inference latency, memory usage, and energy consumption.

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